

SIMATS ENGINEERING

ASSESSMENT TOOL III – ERROR ANALYSIS TASK (LOW)

Course Name: ETHICAL HACKING	Code: CO: CO2
Duration: 90 min	Date: 13-08-2026
Total Marks:	Reg. No / Name : Delhi Ganesh P (192521434d)
CO2: Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks. (BL3)	

Assessment Tool Weightage: 35%

Instructions: Answer all questions. Each question carries 10 marks.

1. A student performs footprinting using Google dorks but constructs queries without operators like “site:”, “filetype:”, or quotation marks. As a result, the search results include unrelated websites and irrelevant documents. The student proceeds with analysis using this incorrect data. Identify the error in query formulation and explain why improper search syntax leads to inaccurate footprinting. Demonstrate how to correct the search query using proper operators and refine results for accurate intelligence gathering.

Answer: The error is using broad Google searches without operators such as site:, filetype:, and quotation marks. This makes the search engine return unrelated websites and documents, resulting in inaccurate footprinting.

Improper syntax reduces the precision of the search because the query does not specify the target domain, exact phrase, or required document type. Therefore, irrelevant information may be mixed with useful intelligence.

Correct examples:

- site:example.com "employee directory"
- site:example.com filetype:pdf "annual report"
- site:example.com filetype:xlsx "contact"

The site: operator restricts results to a particular domain, filetype: searches for a specific

document format, and quotation marks search for an exact phrase. Combining these operators produces more relevant and accurate intelligence.

2. While conducting competitive intelligence, a student gathers company data from blogs and unofficial forums without verifying authenticity. The collected information includes outdated employee details and incorrect organizational structure. Identify the mistake in source validation and explain why unreliable data affects decision-making. Recommend methods to verify information using official sources such as company websites, LinkedIn, and government databases.

Answer: The mistake is accepting information from blogs and unofficial forums without checking whether it is authentic, current, and reliable. This caused outdated employee information and an incorrect organizational structure to be treated as valid.

Unreliable information can lead to incorrect security and business decisions because analysts may identify the wrong employees, departments, technologies, or organizational relationships.

The information should be verified by:

- Checking the company's official website.
- Comparing employee details with LinkedIn profiles.
- Using government databases and official records where available.
- Comparing information from multiple independent sources.
- Checking publication dates and recent updates.

Using trusted and multiple sources improves the accuracy of competitive intelligence.

3. A student directly starts active scanning on a target organization without performing passive footprinting such as WHOIS lookup and search engine analysis. This results in detection alerts being triggered. Identify the mistake in reconnaissance methodology and explain why passive reconnaissance is critical. Suggest the correct sequence of footprinting steps.

Answer: The mistake is starting active scanning before performing passive reconnaissance. Active scanning directly interacts with the target infrastructure and can generate logs, alerts, and intrusion-detection events.

Passive reconnaissance is critical because it collects publicly available information without directly interacting with the target systems. It helps the analyst understand the target before any authorized active testing is performed.

Correct sequence:

1. Define the scope and obtain authorization.

2. Perform passive footprinting using search engines and public sources.
3. Perform WHOIS and DNS analysis.
4. Identify domains, subdomains, IP addresses, and publicly visible services.
5. Analyze and organize the collected information.
6. Perform controlled active scanning only when authorized.
7. Document the findings and assess security risks.

This approach provides better intelligence and reduces unnecessary detection during reconnaissance.

4. A student performs WHOIS lookup but only records domain name and registrar details, ignoring expiry date, administrative contacts, and name servers. Identify the missing components and explain how these fields provide useful security insights. Suggest how proper WHOIS analysis can reveal vulnerabilities.

Answer: The missing components are the domain expiry date, administrative information where publicly available, and name-server details.

- Expiry date can indicate registration and renewal status and may reveal domain-management risks.
- Administrative information, when publicly available, can provide useful information about ownership or administration.
- Name servers reveal the DNS infrastructure responsible for the domain and can indicate relationships with DNS providers.
- Registrar information helps identify where the domain is registered.

Proper WHOIS analysis can provide security insights about domain ownership, registration status, DNS infrastructure, and related organizational information. Analysts can use these details to identify possible weaknesses in domain management and exposed infrastructure, while verifying findings before treating them as vulnerabilities.

5. A student downloads publicly available documents from an organization's website but focuses only on the visible content while ignoring embedded metadata. As a result, important details such as author names, software versions, and internal usernames are missed. Identify the mistake in the student's footprinting process and explain how metadata can provide valuable intelligence. Describe the correct method to extract metadata using appropriate tools. Suggest how this information could be used ethically to identify potential security risks.

Answer: The mistake is examining only the visible document content and ignoring embedded metadata. Metadata can contain information that is not displayed on the document itself.

Useful metadata may include:

- Author or creator names
- Internal usernames
- Software and application versions
- Creation and modification dates
- Organization-related information
- File properties and document history

The correct method is to make a working copy of the authorized public document and inspect its metadata using tools such as ExifTool or document-specific metadata inspection tools. The results should then be recorded and validated against other public information.

Ethically, the information can be used to identify risks such as unnecessary disclosure of internal usernames or software information. These findings should be reported to the organization so that exposed metadata can be removed or minimized.

Code of Conduct:

I **S B DHARSHINI (192465026)** (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **S B DHARSHINI**

RUBRICS FOR CALCULATION

Criteria		Weightage			
Error Identification		20%			
Root Cause Analysis		30%			
Correction & Justification		25%			
Application of Concepts		15%			
Clarity & Presentation		10%			
Performance Levels		Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	
Criteria					
Level 4 –	Exemplary				
Error Identification (2)					
Root Cause Analysis (3)					
Correction & Justification (2)					
Application of Concepts (2)					
Total:	Q1	Q2	Q3	Q4	Q5